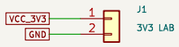
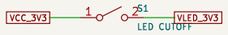
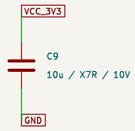
COUPON E0 / 16 x 8 red LEDs / 3.3 V on the bench

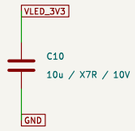
NOT VALIDATED FOR MANUFACTURING — no connection to the vehicle

Current-limited lab power supply. Sources off before wiring.







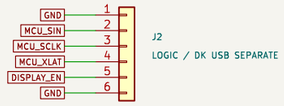


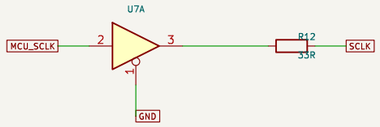


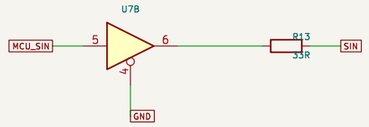


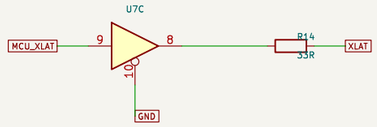


S1 cuts the LED rail; C10 discharges through R15. No hardware watchdog.

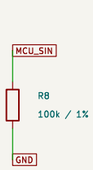


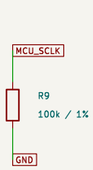


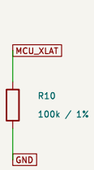




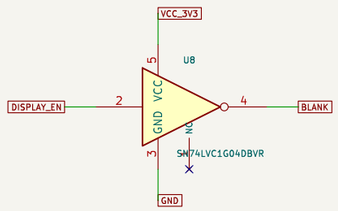
No DK VDD/VBUS on J1/J2. DK GPIO pins to be defined.



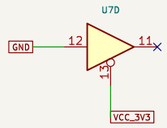


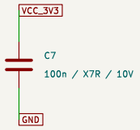


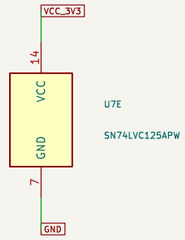


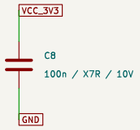












DISPLAY_EN low -> BLANK high -> black. MCU stuck high: use S1.

C7 near U7; C8 near U8. Values and sequences to measure on the bench.

Six drivers TLC5947DAP

File: drivers.kicad_sch

LED matrix 16 x 8

File: leds.kicad_sch

Sheets linked by global labels.
KiCad 9.0.2 symbols; no footprint assigned.

Footprints not assigned; no PCB and no physical validation
NOT VALIDATED FOR MANUFACTURING — 3.3 V BENCH ONLY

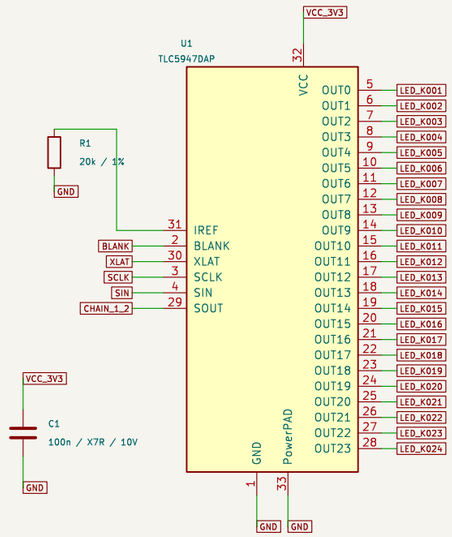
Sheet: /
File: coupon-e0.kicad_sch

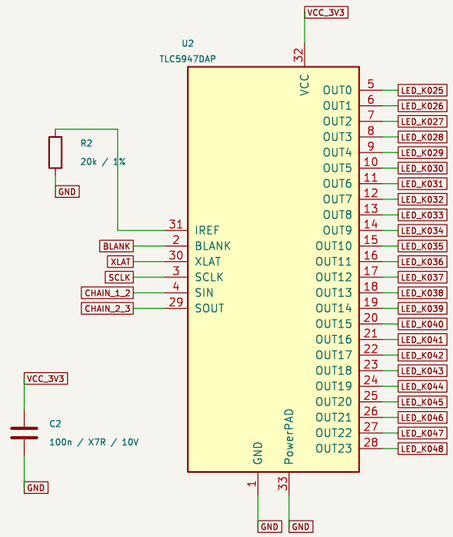
Title: Coupon E0 — Power and control

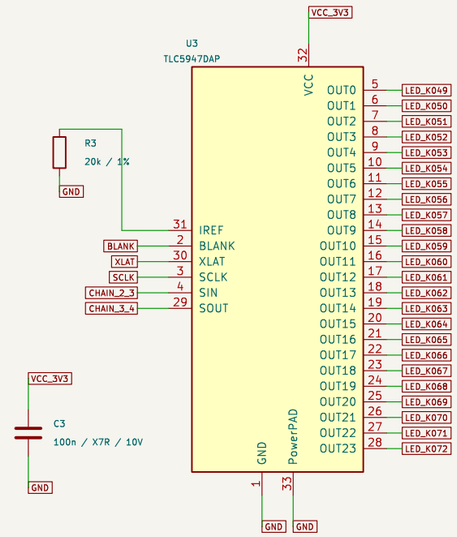
Size: A3	Date: 2026-10-02	**Rev: E0.1**
KiCad E.D.A. 9.0.2		Id: 1/3

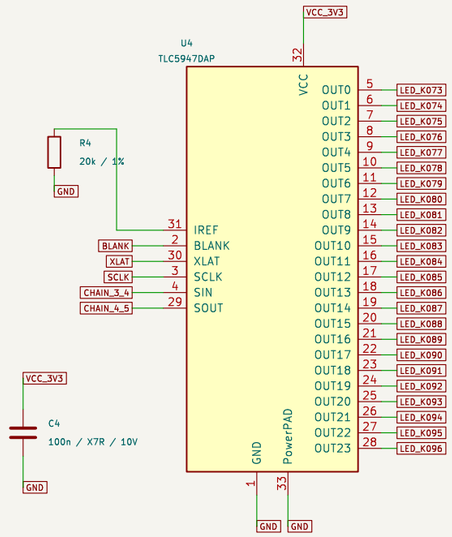
6 x TLC5947DAP / static outputs / chain SIN -> U1 -> ... -> U6

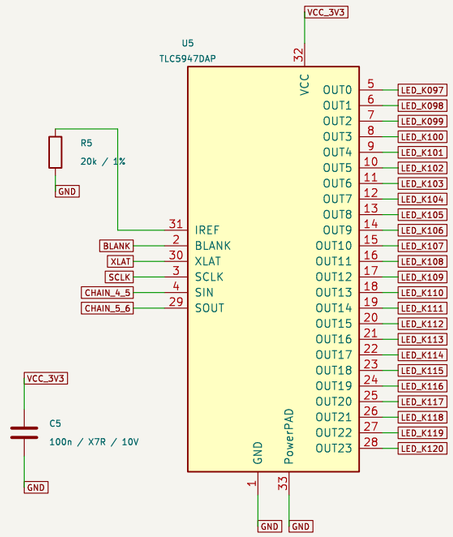
DAP only. Thermal pad = KiCad pin 33, to connect to GND. IREF: 20k 1% / driver.

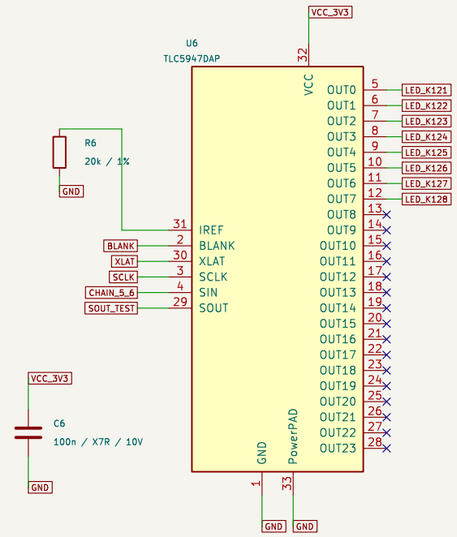




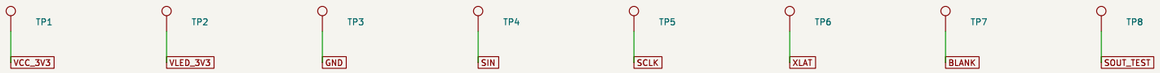








U6 OUT8..23 not connected; PWM data = 0. C1..C6 as close as possible to driver VCC.



Footprints not assigned; no PCB and no physical validation
NOT VALIDATED FOR MANUFACTURING – 3.3 V BENCH ONLY

Sheet: /Six drivers TLC5947DAP/
File: drivers.kicad_sch

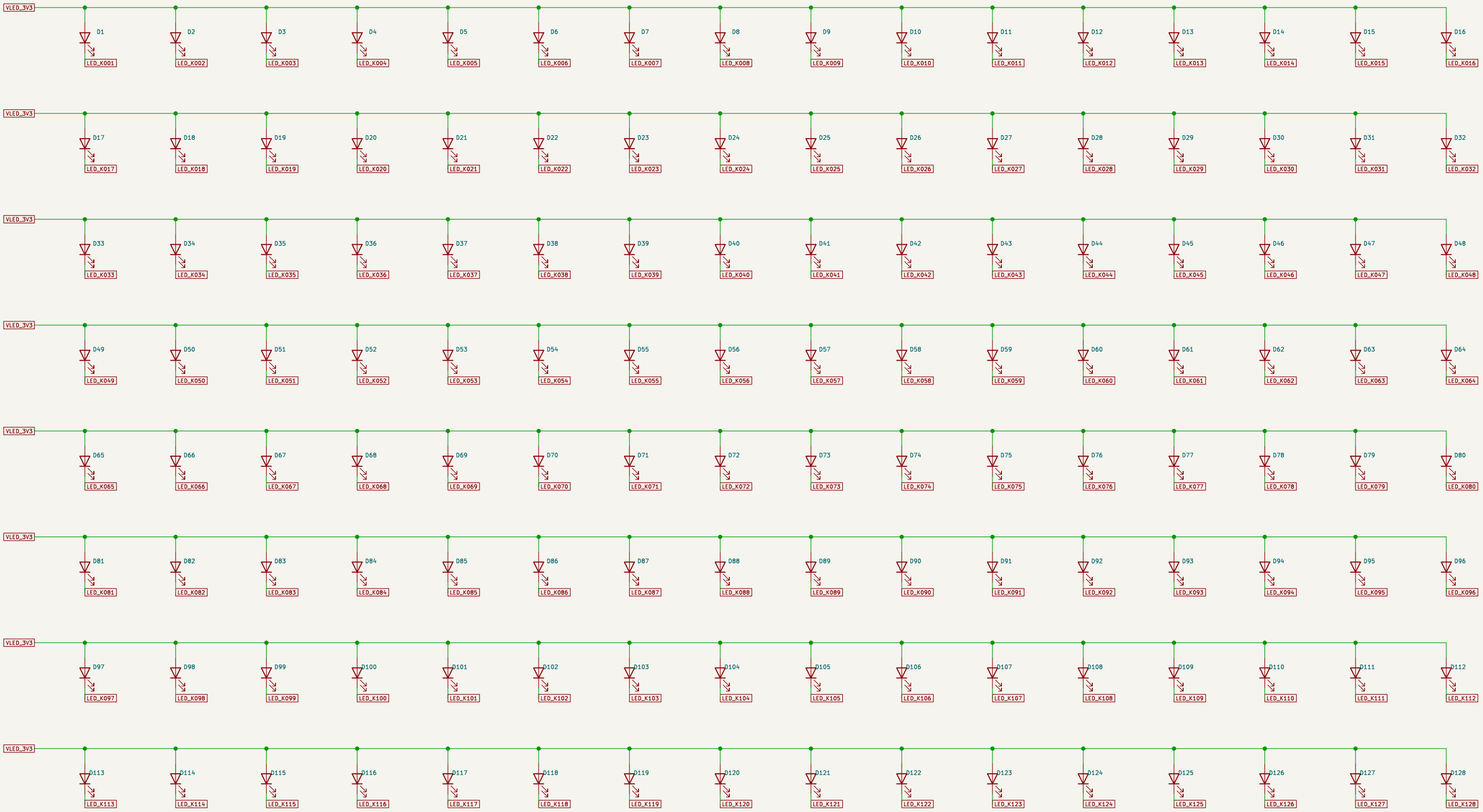
Title: Coupon E0 – Drivers and test points

Size: A3	Date: 2026-10-02	**Rev: E0.1**
KiCad E.D.A. 9.0.2		Id: 2/3

D1..D128: APHHS1005LSECK/J3-PF / anode (2) on VLED_3V3

Logical view of the LED face: rows left to right. Schematic not to scale, no PCB placement.

1 x 0.5 mm package != pitch. Compare 2.5 / 4 mm on the bench. Footprint and polarity to review before routing.



One output per cathode. The 16 remaining U6 outputs are not connected. All footprints remain to be defined.

Footprints not assigned; no PCB and no physical validation
NOT VALIDATED FOR MANUFACTURING – 3.3 V BENCH ONLY

Sheet: /LED matrix 16 x 8/
File: leds.kicad_sch

Title: Coupon E0 – Red matrix 16 x 8

Size: A3	Date: 2026-10-02	**Rev: E0.1**
KiCad E.D.A. 9.0.2		Id: 3/3